



Daffodil International University

Department of Software Engineering
Faculty of Science & Information Technology
Final Examination, Fall 2024
Course Code: ENG 101; Course Title: English-1
Sections & Teachers: All; MA, SSS, SST, AJS

Time: 2:00 Hrs

Marks: 40

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

Read the passage carefully and answer the following questions.

Cognitive Dissonance

- A. Charles Darwin said, "This is not the strongest of the species that survive, nor the most intelligent, but the one most responsive to change." So you've sold your home, quit your job, shunned your colleagues, and abandoned your friends and family. The end of the world is nigh, and you 'know for a fact' that you are one of the chosen few who will be swept up from the 'great flood' approaching on 21st December at midnight to be flown to safety on a far off planet. And then midnight on 21st December comes around and there is no flood. No end of the world. No flying saucer to the rescue. What do you do? Admit you were wrong? Acknowledge that you gave up position, money, friends – for nothing? Tell yourself and others you have been a schmuck? Not on your life.
- B. Social psychologist Leon Festinger infiltrated a flying saucer doomsday cult in the late 1950s. The members of this cult had given up everything on the premise that the world was about to self-destruct and that they, because of their faith, would be the sole survivors. In the lead up to the fateful day, the cult shunned publicity and shied away from journalists. Festinger posed as a cultist and was present when the space ship failed to show up. He was curious about what would happen. How would the disappointed cultists react to the failure of their prophecy? Would they be embarrassed and humiliated? What actually happened amazed him.
- C. However, Festinger coined the term 'cognitive dissonance' to describe the uncomfortable tension we feel when we experience conflicting thoughts or beliefs (cognitions), or engage in behavior that is apparently opposed to our stated beliefs. What is particularly interesting is the lengths to which people will go to reduce the inner tension without accepting that they might, in fact, be wrong. They will accept almost any form of relief, other than admitting being at fault, or mistaken. Festinger quickly realized that our intolerance for 'cognitive dissonance' could explain many mysteries of human behavior.
- D. In a fascinating experiment, Festinger and his colleagues paid some subjects twenty dollars to tell a specific lie, while they paid another group of subjects only one dollar to do the same. Those who were paid just one dollar were far more likely to claim, after the event, that they had actually believed in the lie they were told to tell. Why? Well, because it's just so much harder to justify having done something that conflicts with your own sense of being 'an honest person' for a mere pittance. If you get more money, you can tell yourself: "Yeah, I lied, but I got well paid! It was justified." But for one dollar? That's not a good enough reason to lie, so what you were saying must have been true in the first place, right?
- E. Emotional factors influence how we vote for our politicians much more than our careful and logical appraisal of their policies, according to Drew Westen, a professor of psychiatry and psychology. This may come as little surprise to you, but what about when we learn that our favored politician may be dishonest? Do we take the trouble to really find out what they are supposed to have done, and so possibly have to change our opinions (and our vote), or do we experience that nasty cognitive dissonance and so seek to keep our minds comfortable at the possible cost of truth?

- F. Cognitive dissonance is essentially a matter of commitment to the choices one has made, and the ongoing need to satisfactorily justify that commitment, even in the face of convincing but conflicting evidence. This is why it can take a long time to leave a cult or an abusive relationship - or even to stop smoking. Life's commitments, whether to a job, a social cause, or a romantic partner, require heavy emotional investment, and so carry significant emotional risks. If people didn't keep to their commitments, they would experience uncomfortable emotional tension. In a way, it makes sense that our brains should be hard-wired for monitoring and justifying our choices and actions - so as to avoid too much truth breaking in at once and overwhelming us.

1. a)	Select the most suitable headings for the given paragraphs from the list of ten headings below. Write the appropriate number I-X with paragraph list i-v in your script. There are more headings than the number of paragraphs, so you will not use them all. <i>For example: Paragraph C = IV</i> List of headings: I. Leon Festinger: On Being Stood-up by The Aliens II. Dishonest Politicians? Never! III. Mind Manipulation: The True Reason of Strange Behavior IV. You Can't Handle The Truth! V. The Catastrophe of 21st December VI. Grow Up – Make Cognitive Dissonance Work for You VII. How Many Dollars Would You Take to Tell A Lie? VIII. Revealing Mysteries: Darwin Was Right. IX. Cognitive Dissonance: Who Are You Kidding? X. The High Cost Of Commitment Exposes Us to Cognitive Dissonance 1. Paragraph A _____ 2. Paragraph B _____ 3. Paragraph D _____ 4. Paragraph E _____ 5. Paragraph F _____	[5x1=5]	CLO-3 Level-C4
b)	Identify the correct answer to the following questions. i. Identify the main reason why people fight cognitive dissonance. A. People's unwillingness to accept their mistakes. B. Wish to avoid the awkward feeling of lying for not a good reason. C. A desire to reduce the inner tension. D. Admitting being at fault to survive. ii. Identify the main requirement of commitment to the choices someone has made, and the ongoing need to justify that commitment. A. Emotional tension. B. Sensible reasoning. C. Our brain justifies our choices. D. Heavy emotional investment. iii. Identify the essential quality for survival according to Charles Darwin's quote. A. Responsiveness to change B. Acquiring wealth C. Physical strength D. Intellectual capacity iv. Select the option from below why people believe the lie when they tell during the experiment – A. they were paid more. B. they were paid moderately. C. they were paid less. D. they felt uncomfortable because of lying v. Identify Leon Festinger's curiosity once the prophecy is failed.	[5x1=5]	

A. Media's coverage of the event

C. Scientific explanation for the prophecy

B. The cultists' reaction to disappointment.

D. Financial losses of cult member

Read the passage below and answer the following questions.

The Halifax Explosion

Before the atomic bomb was dropped on Hiroshima in 1945, the largest-ever non-natural explosion had taken place in 1917 in the eastern Canadian port city of Halifax. With the outbreak of World War I, Halifax was effectively transformed into a boomtown. Convoys gathered weekly in Bedford Basin (the north-western end of Halifax Harbor) to traverse the Atlantic, and Halifax Harbor became heavy with vessels of one variety or another. This spike in boat traffic was not dealt with efficiently, and collisions became almost normal.

On December 1, 1917, the French vessel Mont Blanc left New York to join a convoy in Halifax after being loaded with 226,797 kilograms of TNT (an explosive), 223,188 kilograms of benzol (a type of gasoline), 1,602,519 kilograms of wet picric acid (an explosive), and 544,311 kilograms of dry picric acid (another explosive). On December 6, 1917, the Mont Blanc was ushered into Halifax's harbor after the U-boat nets had been raised.

At the same time, the cargo-less Norwegian ship, Imo, left Bedford Basin en route to New York in order to pick up relief items for transport to war-torn Belgium. Imo was behind schedule and attempting to remedy that. She passed a boat on the wrong side before sending a tugboat retreating to port. By the time she reached the Narrows, she was in the wrong channel and going too fast. The Mont Blanc sounded her whistle, but the Imo sounded back twice, refusing to alter course. At the last moment, the Mont Blanc veered, and the Imo reversed, but it was too late. From the gash formed in the French boat's hull seeped a noxious spiral of oily, orange-dappled smoke. Mont Blanc's crew rowed to shore on the Dartmouth side, but no one could decipher their warnings. Their fiery vessel then casually drifted toward the Halifax side where it came to rest against one of the piers.

The Halifax Explosion, as it became known, was the largest manmade detonation to date, approximately one-fifth the ferocity of the bomb later dropped on Hiroshima. It sent up a column of smoke reckoned to be 7,000 meters in height. It was felt more than 480 kilometers away. It flung a ship gun barrel some 5.5 kilometers, and part of an anchor, which weighed 517 kilograms, around 3 kilometers. The blast absolutely flattened a district known as Richmond. It also caused a tsunami that saw a wave 18 meters above the high-water mark deposit the Imo onto the shore of the Dartmouth side. The pressure wave of air that was produced snapped trees, bent iron rails, and grounded ships. That evening, a blizzard commenced, and it would continue until the next day, leaving 40 centimeters of snow in its wake. Consequently, many of those trapped within collapsed structures died of exposure. Historians put the death toll of the Halifax Explosion at approximately 2,000.

c) Trace the answer to the following questions. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- i. Identify what the Mont Blanc was carrying apart from explosives.
- ii. Point out the part of the Mont Blanc that was damaged due to the collision.
- iii. Identify the location to date where the most massive human-made explosion has taken place.
- iv. Select the place that was entirely devastated by the explosion.
- v. Point out the elements that threw the Imo off to the shore.

[5x1=5]

2.	a) Write the correct form of the verbs in the bracket to complete the following conditional sentences as directed. - You ____ (get) water if you ____ (mix) hydrogen and oxygen. (Use Zero Conditional) - She ____ (travel) all over the world if she ____ (be) rich. (Use Second Conditional) - She ____ (miss) the bus if she ____ (not leave) soon. (Use First Conditional) - If we ____ (take) a taxi, we ____ (not miss) the plane. (Use Third Conditional) - If you ____ (touch) a fire, you ____ (get) burned. (Use Zero Conditional) - If the train ____ (be) delayed, she ____ (be) late to attend the meeting. (Use First Conditional)	[6x0.5= 3]	CLO-1 Level-C1
	b) Identify if the following sentences are Fragment, Run-on, Modifier, or Complete. If the sentence is complete, write "Complete Sentence"; if the sentence is Fragment, Run-on, or Modifier, then rewrite the sentence to fix the syntax error. - Our company needs to address the client's complaints we cannot afford to lose their trust.↵ - Close to the mountain.↵ - She almost drove her kids to school every morning.↵ - As he was suffering from depression.↵ ✓ The experiment failed to produce the expected results the team is now considering alternative hypotheses they hope to achieve success in the next phase.↵ - After a long day at work, I went home, made a simple dinner, and settled onto the couch with a good book, determined to unwind and recharge for the next day.↵ ✓ Hanging in the hallway, the students admired the new painting.↵	[7x1=7]	
3.	a) You are a final-year student at Basel University, studying Software Engineering. You want to apply for the post of "Teaching Assistant" you have recently seen on your department's notice board. Compose a letter to the HOD. In your letter, * Give your reason for writing * Tell about your education and describe your work experience * Explain why you would be suitable for the position	[7]	C Le
	b) Construct an essay on any one of the following topics: - Nowadays university education is considered very important for people's future. However, there are a lot of successful people who didn't get higher education. Do you think that higher education is necessary to succeed in life? - Deforestation is the large-scale clearing of forests, leading to biodiversity loss, climate change, and disrupted ecosystems. What do you think are the causes and effects of deforestation? What solutions can you offer?	[18]	



Daffodil International University

Department of Software Engineering
Faculty of Science & Information Technology

Final Examination, Fall 2024

Course Code: SE 111; Course Title: Computer Fundamentals

Sections & Teachers: All

Time: 2:00 Hrs

Marks: 40

Answer **ALL** Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Imagine you are a healthcare assistant at a wellness center. Your task is to help classify people based on their Body Mass Index (BMI), which is calculated using the formula: $\text{BMI} = (\text{Height (m)})^2 / \text{Weight (kg)}$ Based on their BMI, categorize them into one of the following health zones: Underweight: $\text{BMI} < 18.5$ Healthy Weight: $18.5 \leq \text{BMI} \leq 24.9$ Overweight: $25 \leq \text{BMI} \leq 29.9$ Obese: $\text{BMI} \geq 30$ For classifying the person into the appropriate health zone illustrate an algorithm and draw flowchart to explain the process.	13	CLO-3 C3
	b)	Detect errors in the provided C code below, update the code after making the necessary corrections, and write down the updated code. Additionally, show the expected output when the input is: 15 10 Code: <pre>#include <studio.h> INT main() { int 1_num; print("Enter a number:) , scan("d", num) if (num % 3 = 0 num % 5 = 0); { prnt("The number is divisible by both 3 and 5\nm) } else { prnt("The number is not divisible by both 3 and 5\nm) } ret 0; }</pre>	7	

c)	Table: Student <table border="1"> <thead> <tr> <th>StudentID</th> <th>Name</th> <th>Age</th> <th>CGPA</th> <th>CourseID</th> </tr> </thead> <tbody> <tr> <td>025</td> <td>Alice</td> <td>22</td> <td>3.50</td> <td>101</td> </tr> <tr> <td>059</td> <td>Bob</td> <td>23</td> <td>3.65</td> <td>102</td> </tr> <tr> <td>009</td> <td>Charlie</td> <td>22</td> <td>3.45</td> <td>103</td> </tr> <tr> <td>095</td> <td>Mark</td> <td>24</td> <td>3.85</td> <td>101</td> </tr> </tbody> </table> Table: Course <table border="1"> <thead> <tr> <th>CourseID</th> <th>CourseName</th> <th>Credit</th> </tr> </thead> <tbody> <tr> <td>101</td> <td>Mathematics</td> <td>3</td> </tr> <tr> <td>102</td> <td>Physics</td> <td>1</td> </tr> <tr> <td>103</td> <td>English</td> <td>2</td> </tr> <tr> <td>105</td> <td>Programming</td> <td>4</td> </tr> </tbody> </table> In the context of databases, two of the most significant keys are present. These keys are essential for ensuring data integrity and establishing relationships between tables. Detect the two keys from the above tables and demonstrate the difference between these two keys.	StudentID	Name	Age	CGPA	CourseID	025	Alice	22	3.50	101	059	Bob	23	3.65	102	009	Charlie	22	3.45	103	095	Mark	24	3.85	101	CourseID	CourseName	Credit	101	Mathematics	3	102	Physics	1	103	English	2	105	Programming	4	5	
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105	Programming	4																																									
d)	Daffodil International University (DIU) has set up two different networks for its departments. Scenario 1: In the Software Engineering Department, all devices connect to a central hub, which also has a backup hub for redundancy. If both hubs fail, the network stops functioning. Scenario 2: In the Computer Science Department, each device is connected to every other device. This setup ensures communication continues even if one connection fails, but managing the connections is increasingly difficult as devices are added. Determine the network topology based on the given description, provide your reasoning and a diagram for each topology.	5																																									
2.	a) Story 1: An online health blog lets users leave comments on articles. A visitor writes a comment containing <code><script>alert('Hacked!');</script></code>. When others view the page, the script runs and shows the alert. In some cases, it also steals login session cookies. Story 2: A hospital's patient portal allows users to search for patient records by entering a patient ID. A user discovers that entering special characters instead of a patient ID, like <code>101'; DROP TABLE patients;--</code>, causes the system to crash and deletes all patient data in the database. Read the two stories, and identify the software vulnerability mentioned in the scenario and explain these briefly.	5	CLO-3 C4																																								
b)	A hospital launches a secure telemedicine platform for patient-doctor consultations. Patient records are stored and sent in a safe format that only authorized doctors can read. Only verified users can access the system, and even if someone intercepts the data, it stays safe and untampered. This technology transformed information in a way that, ideally, only authorized parties can decode. Identify the technology mentioned in the scenario step by step and explain it with proper explanation with appropriate diagram.	5																																									



Daffodil International University

Faculty of Science & Information Technology

Department of Software Engineering

Final term Examination, Fall 2024

Course Code: SE 113; Course Title: Introduction to Software Engineering

Sections & Teachers: All & AP, DAKM, MKS, JIC, TH, SAN, MIM, ST

Time: 2 Hours

Marks: 40

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Suppose, you are designing an <u>online library system</u> that allows users to <u>search</u> for <u>books</u> , <u>borrow them</u> , and <u>view their borrowing history</u> . Describe the activities involved in the design phase. <i>Include examples</i> of outputs for each activity.	[Marks-5]	CLO-1 Level-2
	b)	Why do we need <u>software testing</u> ? Describe the <u>role of a software tester</u> in the software development process.	[Marks-5]	CLO-1 Level-2
2.	a)	In the One Card app, users must register, log in to access their account, and log out after completing their transactions. You are tasked with creating a <u>use case diagram</u> for these functionalities. Illustrate the use case diagram and <i>write down</i> the description for the Register, Login, and Logout functionalities in the <u>Bkash app</u> .	[Marks-5]	CLO-2 Level-3
	b)	<u>Demonstrate</u> different techniques of Black-box testing and <i>provide an example</i> for each.	[Marks-5]	CLO-2 Level-3
	c)	Define software security. Generalize the steps in implementing software security in software development industry.	[Marks-5]	CLO-2 Level-3
3	a)	You are part of a team developing a one-card application. During the project, the following challenges arise: i. The project is delayed because the team underestimated the time required for testing, leading to a missed deadline. ii. Midway through development, the client requests significant changes to the app's features, increasing the workload. iii. The <u>programming language chosen</u> for the app has limited community support, resulting in integration challenges with the existing database system. iv. During testing, the app performs poorly under heavy user loads, with slow <u>response times</u> . v. The team struggles with communication, leading to repeated misunderstandings about feature requirements. <u>Classify the types of software risks</u> involved in this scenario and <i>describe</i> each challenge accordingly.	[Marks-5]	CLO-3 Level 2

b)	Considering the project information presented in the table below of Critical Path Analysis <table><tr><th>Activity</th><th>Immediate Predecessor</th><th>Time (days)</th></tr><tr><td>A</td><td>-</td><td>2</td></tr><tr><td>B</td><td>-</td><td>5</td></tr><tr><td>C</td><td>-</td><td>1</td></tr><tr><td>D</td><td>B</td><td>10</td></tr><tr><td>E</td><td>A, D</td><td>3</td></tr><tr><td>F</td><td>C</td><td>6</td></tr><tr><td>G</td><td>E, F</td><td>8</td></tr></table> Visualize the network diagram from the above table. Also, estimate all the possible path and their duration and compute the critical path of the project.	Activity	Immediate Predecessor	Time (days)	A	-	2	B	-	5	C	-	1	D	B	10	E	A, D	3	F	C	6	G	E, F	8	[Marks-6]	CLO-3 Level-2
Activity	Immediate Predecessor	Time (days)																									
A	-	2																									
B	-	5																									
C	-	1																									
D	B	10																									
E	A, D	3																									
F	C	6																									
G	E, F	8																									
c)	Assume that the expected size of a semi-detached software project is 250.3 KLOC. Using the coefficients ($a_b = 3$, $b_b = 1.12$, $C_b = 2.5$, $d_b = 0.35$), the BASIC COCOMO MODEL, estimate the recommended number of people who need to be engaged to complete the project.	[Marks-4]	CLO-3 Level-2																								



Daffodil International University

Department of Software Engineering
Faculty of Science & Information Technology

Final Examination, Fall 2024

Course Code: MAT 101 ; Course Title: Mathematics I

Sections & Teachers: All

Time: 2:00 Hrs

Marks: 40

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Describe inflection point with example.	[Marks-01]	CLO-1 Level-2
	b)	Classify the maxima-minima of the function: $f(x) = \frac{1}{6}x^6 - 12x^5 + 36x^4 + 6$	[Marks-06]	
2.	a)	If $u = \ln\sqrt{(x^2 + y^2 + z^2)}$, then express $(x^2 + y^2 + z^2) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = 1$.	[Marks-04]	CLO-1 Level-2
	b)	Explain Euler's theorem. Also using it express $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{4} \sin 2u$, when $u = \tan^{-1} \left(\frac{x+y}{\sqrt{x+y}} \right)$.	[Marks-04]	
3.	a)	Solve the following integrals: i. $\int \cos x \cos 4x \cos 5x \, dx$ ii. $\int \frac{\tan x \sec^2 x}{a^2 + b^2 \tan^2 x} \, dx$ iii. $\int (\sin^{-1} x)^2 \, dx$ iv. $\int_0^{\pi} \frac{dx}{2 + \cos x}$	[Marks-2.5×4=10]	CLO-2 Level-3
4.	a)	Examine the area of the circle: $x^2 + y^2 = 16$.	[Marks-05]	CLO-2 Level-3
5.	a)	Using Gamma-Beta function analyze the followings: i. $\int_0^1 x^{\frac{5}{2}} (1-x)^{\frac{3}{2}} \, dx$ ii. $\int_0^{\frac{\pi}{2}} \sin 2x \cos^6 x \, dx$	[Marks-2×5=10]	CLO-3 Level-4



Daffodil International University
Department of Software Engineering
Faculty of Science & Information Technology
Final Examination, Fall-2024
Course Code: BNS-101, Course Title: Bangladesh Studies
Sections: 43_All Teacher Initials: MBU, SHS, MSI

Time: 2 Hours

Marks: 40

Answer ALL the Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	Illustrate what strategies should initiate in Bangladesh's ICT industry for addressing the problems to establish itself as a premier center for creative solution hub.	08	CLO-5 Level 3
2.	Identify the basic initiatives of environmental ethics in defining global accountability for environmental protection. As an engineer, what would you suggest in your workplace regarding sustainable eco-friendly domain.	08	CLO-4 Level 4
3.	Determine how can Bangladesh shift from brown economy to green economy by implementing the 3R principles of the circular economy, to achieve sustainable economic growth.	08	CLO- Level
4.	"Urbanization as a social trend"- Explain the role of push & pull factors for urbanization process with exploring the implementation of rural development challenges.	08	CLO Lev
5.	Distinguish the spirit of the Liberation War in 1971 compared with the goals and challenges of the July Revolution_2024 in changing the national identity and people's hopes.	08	CL Le